ML · #11

Predicting Metabolic Cost During Human-in-the-Loop Optimisation

Replacing a slow, lab-only face-mask measurement with wearable EMG and gait sensors, for exoskeletons that tune themselves outside the lab

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UE24CS352A Machine Learning · Mini-Project Review · October 2026

THE PROBLEM

Exoskeleton tuning is bottlenecked by a breathing mask

CMA-ES optimiser
proposes new torque profile

→

Exoskeleton
applies it for 2 min

→

Human walks
body adapts

→

Metabolic cost
measured via face mask

↻ *repeated 36× per 72-min session, feedback = metabolic cost*

~1 min
to settle after every change in control law, so it is slow feedback

≈ 5 %
irreducible noise in the measurement (MSE floor ≈ 0.0025)

Lab only
needs a face mask, a metabolic cart and fasting subjects

Our goal: learn f(EMG, gait, controls) → metabolic cost so the mask can eventually be replaced by sensors a person could wear anywhere.

DATASET

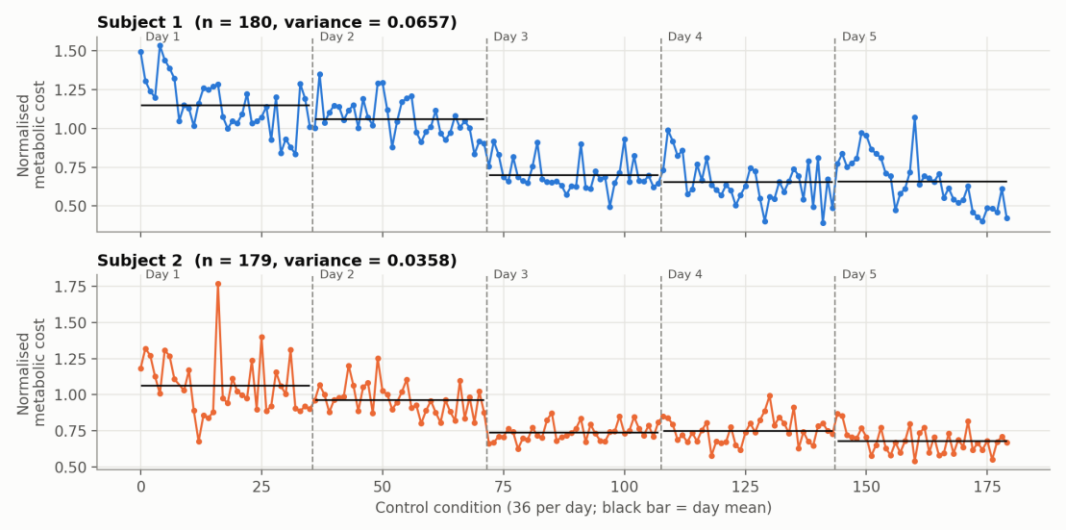
Real lab data: 2 people, 5 days, 36 assistance conditions per day

359
samples
(180 + 179)

29
features per
sample

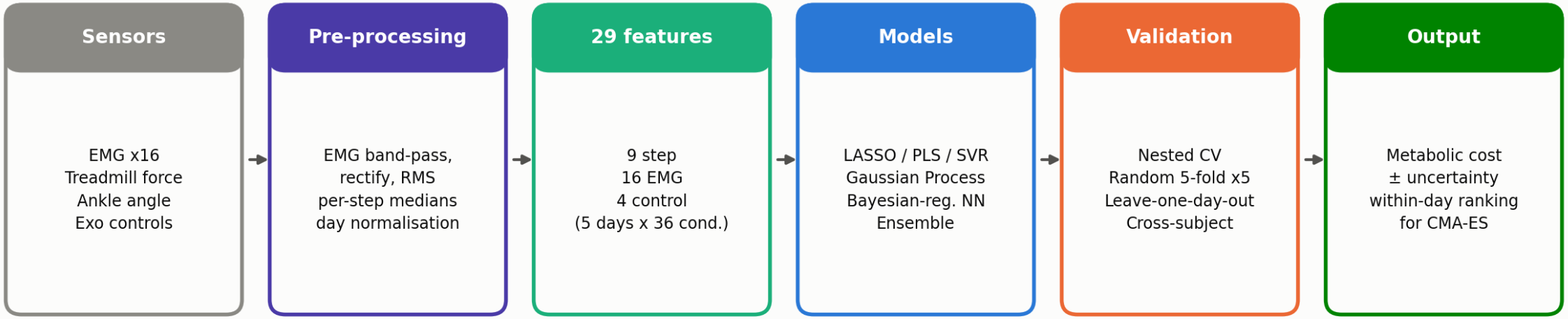
0.0025
noise-floor
MSE

- **16 EMG**
  8 muscles per leg; band-pass → rectify → RMS, normalised each day
- **9 Step / gait**
  peak ground force, ankle dorsi/plantar-flexion, step time, step width
- **4 Control**
  exo torque profile: peak torque, peak time, rise time, fall time
- **1 Target**
  metabolic rate - standing baseline, ÷ normal walking (1.0 = no exo)



METHODOLOGY

End-to-end pipeline



1 Leakage-free

Scaling and every hyper-parameter search happen inside training folds (nested CV). The paper tuned λ on its test folds.

2 Two protocols

Random 5-fold ×5 (the paper's, for comparison) and leave-one-day-out (predict a whole new session, the realistic case).

3 Two layers

A faithful 1:1 Python port of the paper's MATLAB scripts (baseline), plus the extended study: 12 experiments, 16 figures, 10 unit tests.

MODELS

13 models, including the paper's network rebuilt from scratch

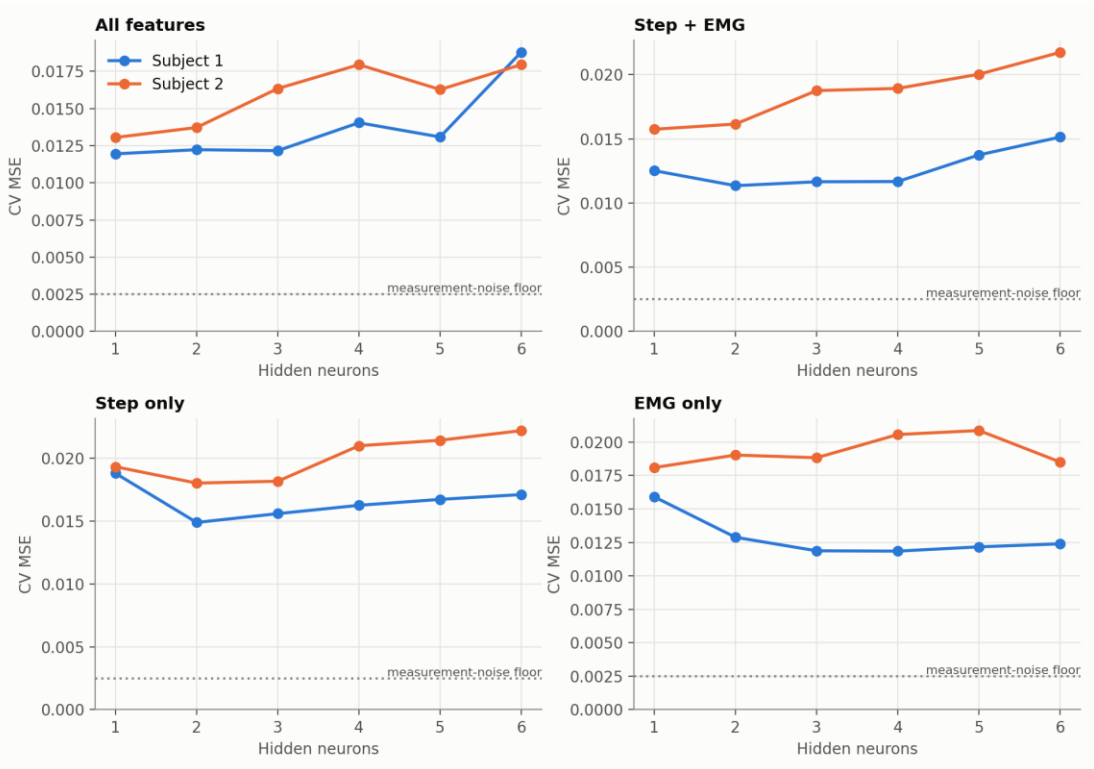
Linear	OLS · LASSO · Ridge · Elastic Net · PLS
Kernel / instance	SVR (RBF) · kNN · Gaussian Process
Trees	Random Forest · Extra Trees · Gradient Boosting
Neural	Bayesian-regularised NN (trainbr)
Ensemble	average of GP + SVR + LASSO + Extra Trees

Bayesian-regularised NN, from scratch

$F(w) = \beta \cdot E_D + \alpha \cdot E_W$

$\gamma = P - \alpha \cdot \mathrm{tr}(H^{-1}),\ \alpha = \gamma / 2E_W,\ \beta = (N - \gamma) / 2E_D$

Levenberg-Marquardt · closed-form Jacobian · picks 1-2 neurons



RESULTS

We beat the original study on both subjects

0.0079

Subject 1 MSE (ours)
paper: 0.0089

−11 %

error vs paper
on Subject 1

0.0109

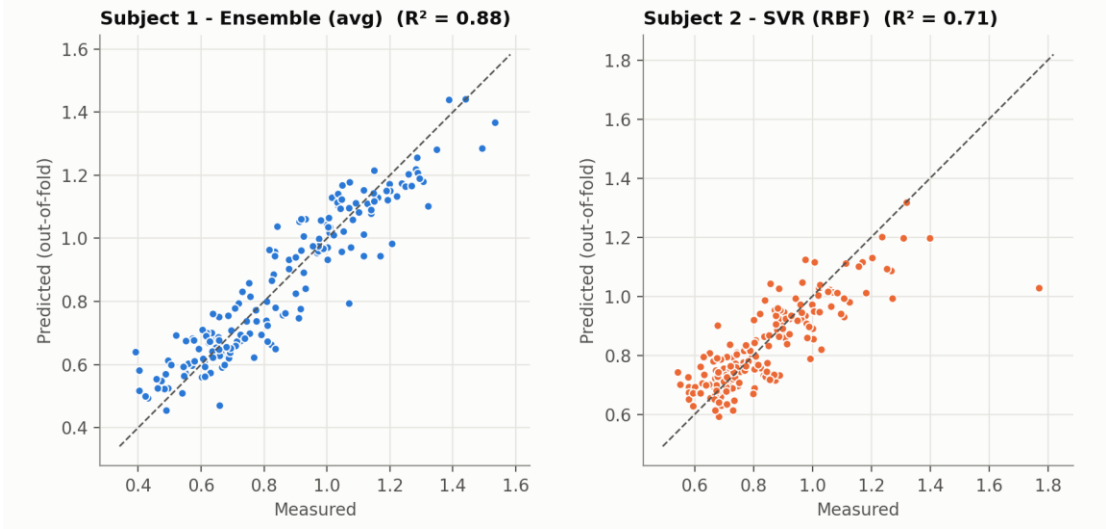
Subject 2 MSE (ours)
paper: 0.0168

−35 %

error vs paper
on Subject 2

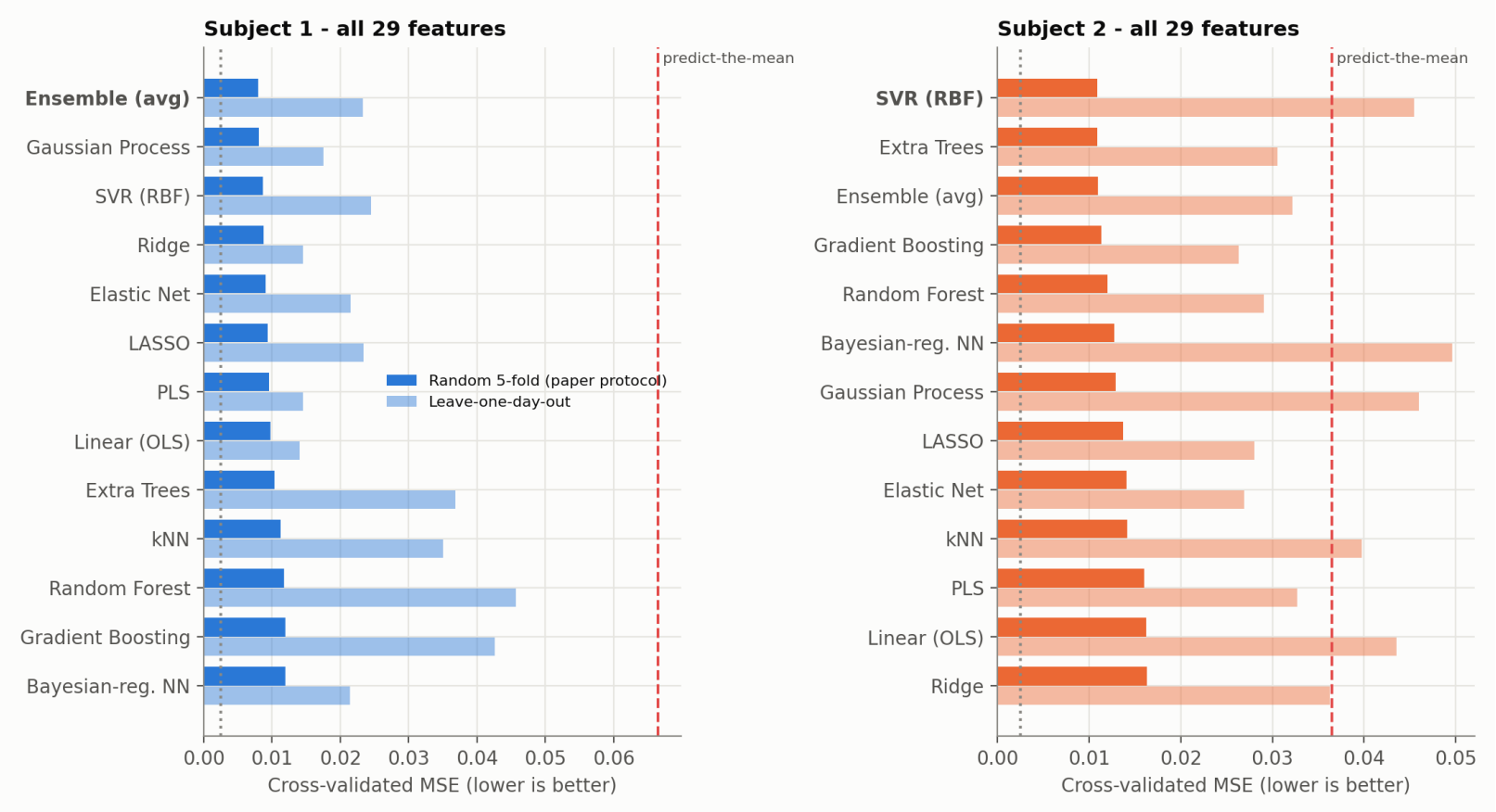
R^2 = 0.88 / 0.71

variance explained (S1 / S2), out-of-fold



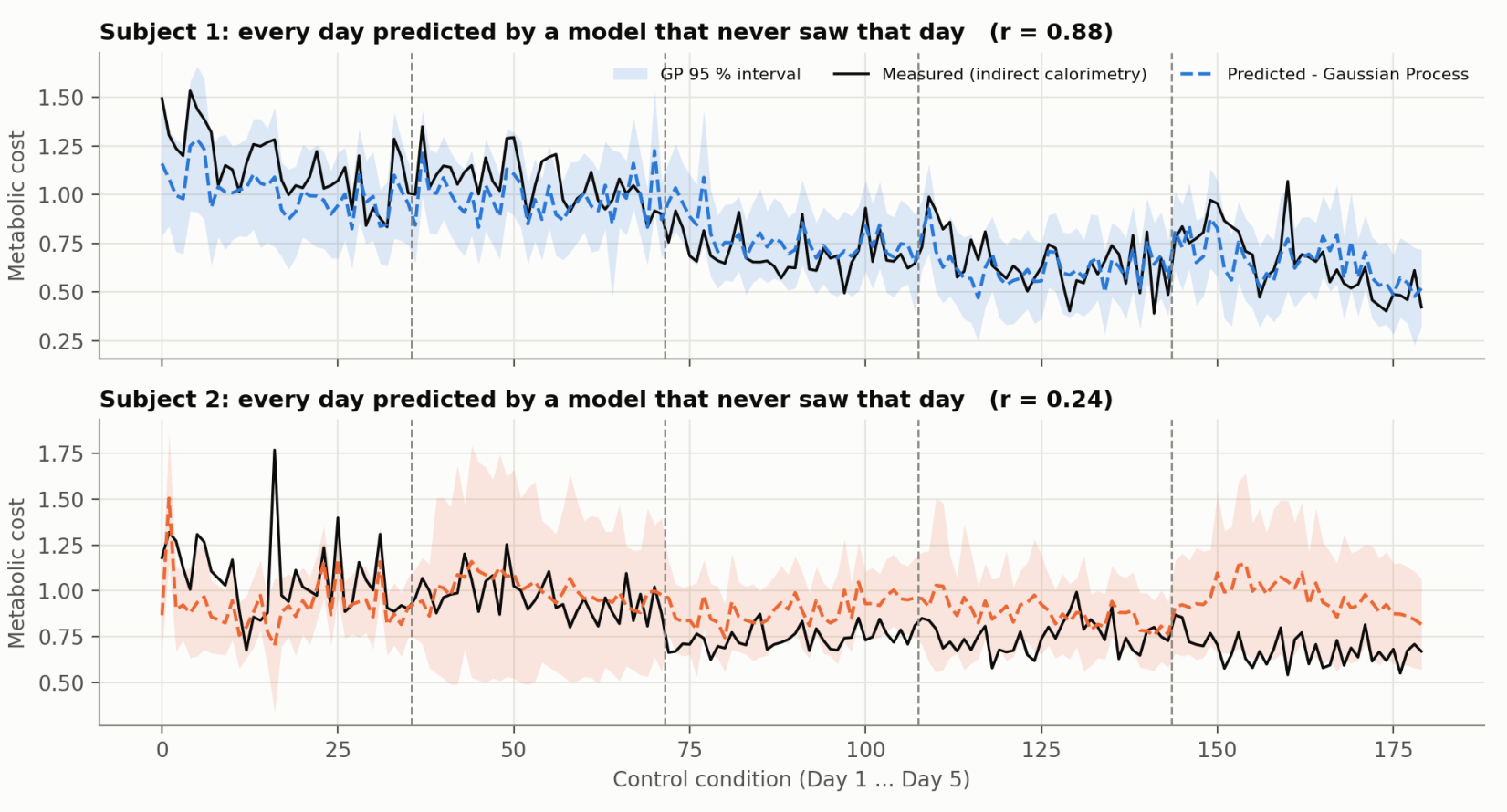
RESULTS · VALIDATION

Random splits flatter models: unseen days are the real test



- **Random CV:** GP, SVR and the ensemble lead (MSE ≈ 0.008-0.011)
- **Unseen day:** error roughly doubles
- **Simple wins across days:** PLS / Ridge reach 0.0145 on S1
- **Trees & kNN** overfit within-day structure
- **Lesson:** report the protocol that matches deployment

Predicting a session the model has never seen



≈ 12 %

error on unseen days, Subject 1 (R^2 0.78)

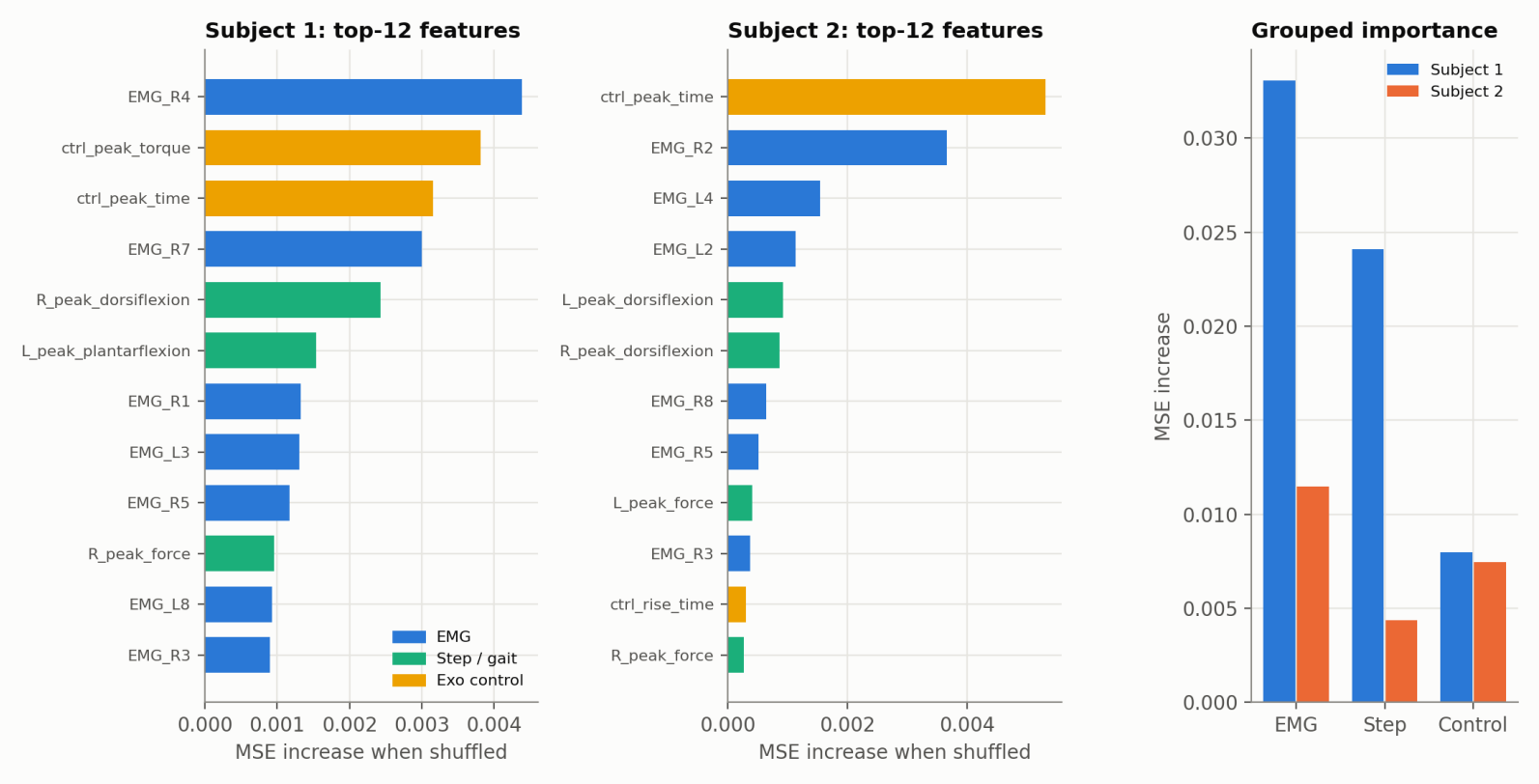
ρ ≈ 0.62

within-day rank correlation (S1), what the optimiser needs

≈ 16 %

Subject 2: day-level shifts are invisible to the sensors

EMG carries the signal, and a few muscles do most of the work

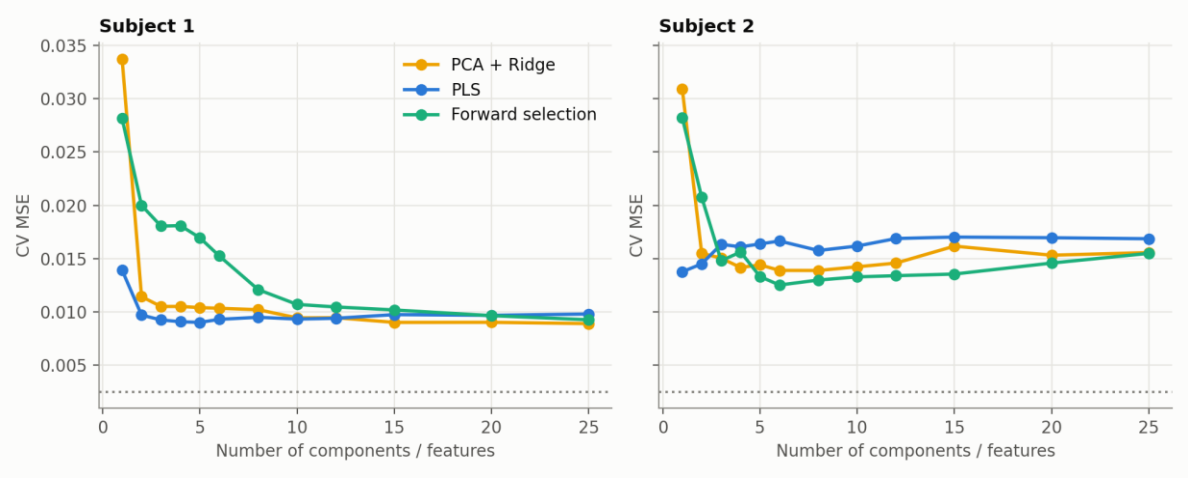


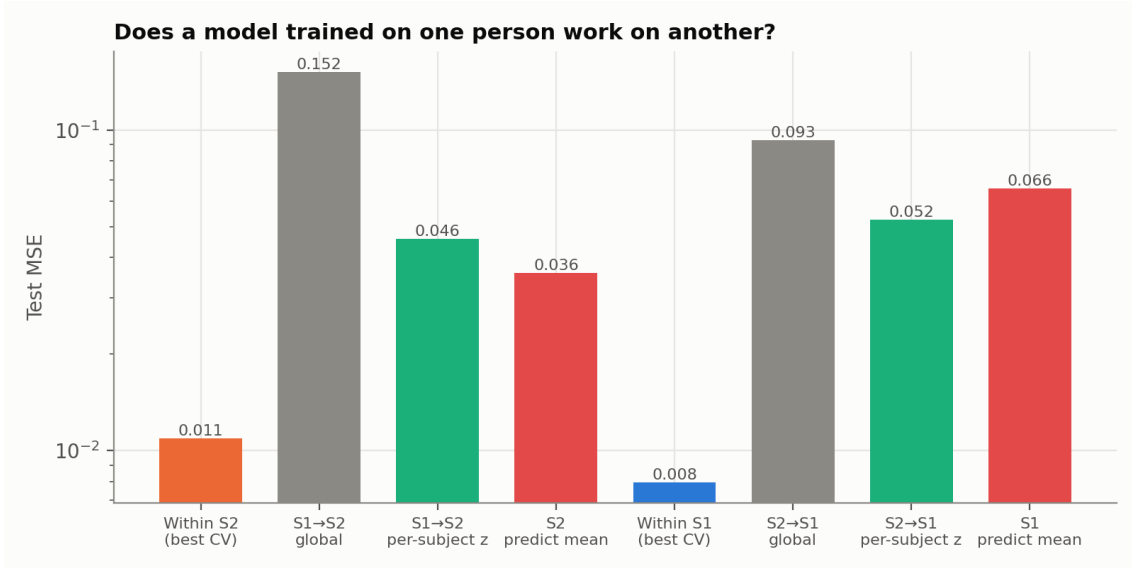
Feature set	S1 GP	S2 GP
All features	0.0081	0.0129
Step + EMG	0.0090	0.0120
EMG only	0.0107	0.0146
Step only	0.0151	0.0141

Feature-set ablation (CV MSE)

- **EMG is the strongest group** in grouped importance for both subjects; dropping the controls costs little, which helps transfer to other exo hardware

Why PCA failed, and why one model cannot fit everyone





Supervised > unsupervised

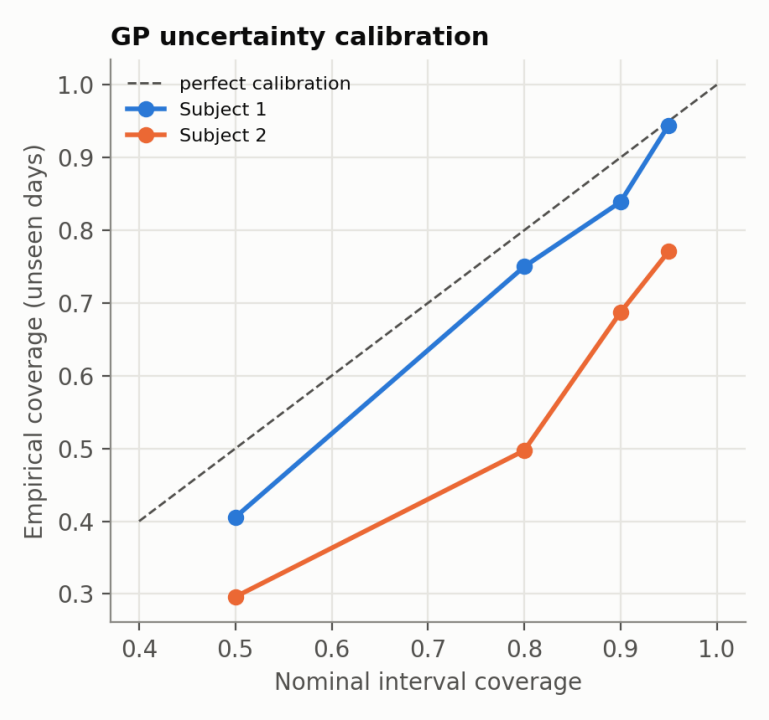
1 PLS component (0.0138) = 8 PCA components on S2. Forward selection is greedy and unstable.

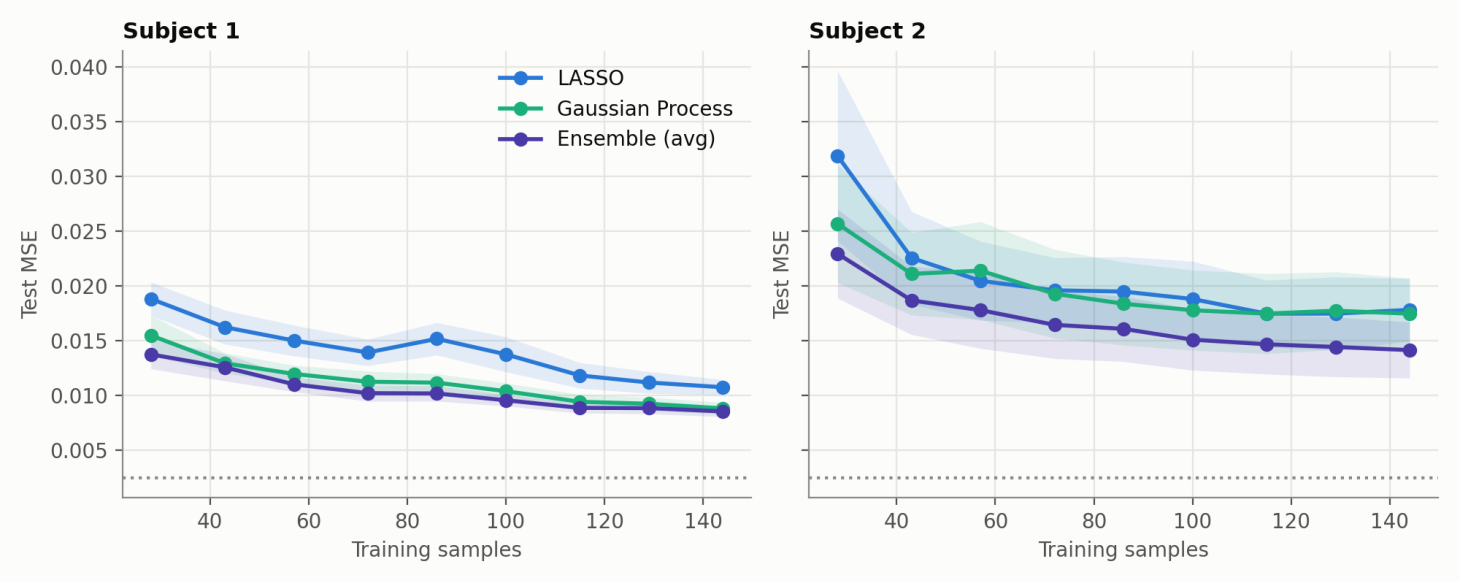
People are different

S1 → S2: MSE 0.152 vs 0.036 for the mean. Per-subject z-scoring → 0.046. Personal models are needed.

RESULTS · RELIABILITY

The model knows when it is unsure, and more data would help





94 %

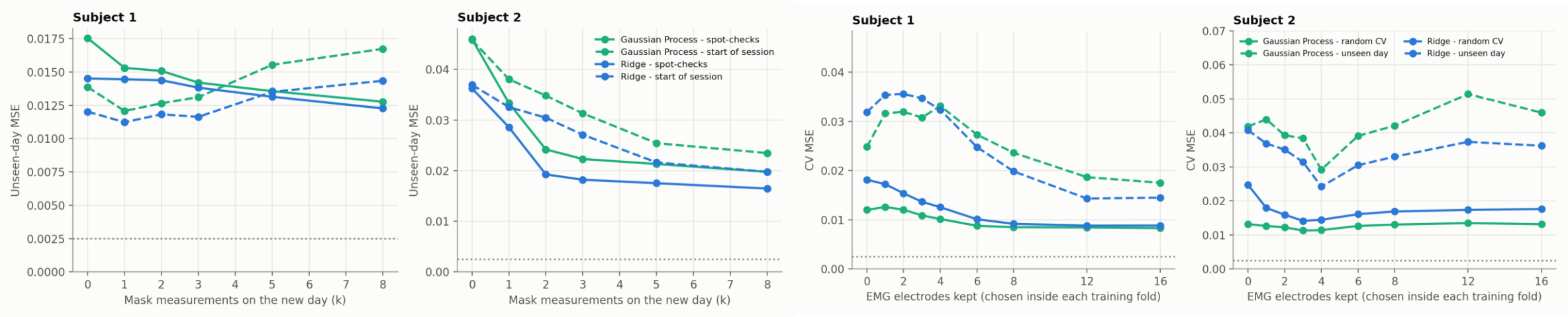
of unseen-day points inside the S1 95 % interval

0.0155 → 0.0088

S1 GP test MSE from 28 to 144 training samples, still decreasing

IMPROVEMENTS

Fixing the weaknesses, without collecting new data



−51 %

S2 unseen-day error with just 3 of 36 mask spot-checks (GP 0.046 → 0.022)

$$w = k/(k+\sigma^2/\tau^2)$$

empirical-Bayes shrinkage: trusts calibration only as much as the training days justify

4 of 16

EMG electrodes beat all 16 for S2 (unseen-day Ridge 0.036 → 0.024)

DEMO

Live demo: the code, running

1 Paper reproduction

```
$ python
paper_reproduction/lin_reg_test.py


Linear Regression 5-Fold CV
Mean MSE: 0.010700
Mean R²  : 0.8349
```

2 Predict unseen days

```
$ python scripts/predict.py --demo
--subject S1


day      MSE    %err      r
  1   0.0286  16.9%   0.82
  2   0.0191  13.8%   0.49
  3   0.0132  11.5%   0.64
  4   0.0119  10.9%   0.61
  5   0.0149  12.2%   0.67
```

3 Automated tests

```
$ python -m pytest -q



.........
10 passed
```

Run from the project folder: cd "ML proj" · total time under a minute

CONCLUSIONS & FUTURE WORK

Wearables can replace the mask, one person at a time

9-10 % error within a session from EMG + gait (mean-baseline: 19-26 %); beats the original study

2× error on a new day: random-split CV is optimistic, so deploy-style validation matters

EMG is the key signal; 4 of 16 electrodes beat all 16 for S2, so cheaper hardware is possible

1 : 1 models must be subject-specific; cross-person transfer is worse than the mean

Next steps

- More subjects + anthropometrics + IMU kinematics
- Test spot-check calibration live in a real HILO session
- Use the surrogate as a prior inside CMA-ES / Bayesian optimisation

Thank you, questions?

Code, data, figures and the 2-page write-up are in the private GitHub repository.

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Ref: Krimsky & Ng, CS229 2018 · Zhang et al., Science 2017 · MacKay 1992 · Rasmussen & Williams 2006